Stat 145 (Multivariate Statistics)

Lesson 3.1 Principal Components Analysis

Learning Outcomes

1. Discuss applications of principal components analysis.
2. Derive principal components.
3. Visualize and interpret principal components.

Introduction

A principal component analysis is concerned with explaining the variance–covariance structure of a set of variables through a few linear combinations of these variables. Its general objectives are (1) data reduction and (2) interpretation. Although p components are required to reproduce the total system variability, often much of this variability can be accounted for by a small number k of the principal components. If so, there is (almost) as much information in the k components as there is in the original p variables.The k principal components can then replace the initial p variables, and the original data set, consisting of n measurements on p variables, is reduced to a data set consisting of n measurements on k principal components.

Analyses of principal components are more of a means to an end rather than an end in themselves, because they frequently serve as intermediate steps in much larger investigations. For example, principal components may be inputs to a multiple regression or cluster analysis.

Population Principal Components

Algebraically, principal components are particular linear combinations of the p random variables Geometrically, these linear combinations represent the selection of a new coordinate system obtained by rotating the original system with $X_1, X_2, \cdots, X_p$ as the coordinate axes. The new axes represent the directions with maximum variability and provide a simpler and more parsimonious description of the covariance structure.

Principal components depend solely on the covariance matrix $\mathbf{\Sigma}$ (or the correlation matrix $\boldsymbol{\rho}$) of $X_1, X_2, \cdots, X_p$. Their development does not require a multivariate normal assumption. On the other hand, principal components derived for multivariate normal populations have useful interpretations in terms of the constant density ellipsoids. Further, inferences can be made from the sample components when the population is multivariate normal.

Let the random vector $\mathbf{X}' = [X_1, X_2, \cdots, X_p]$ have the covariance matrix $\mathbf{\Sigma}$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$. Consider the linear combinations

$$\begin{aligned}
Y_1 &= \mathbf{a}_1'\mathbf{X} = a_{11}X_1 + a_{12}X_2 + \cdots + a_{1p}X_p \\
Y_2 &= \mathbf{a}_2'\mathbf{X} = a_{21}X_1 + a_{22}X_2 + \cdots + a_{2p}X_p \\
&\vdots \\
Y_p &= \mathbf{a}_p'\mathbf{X} = a_{p1}X_1 + a_{p2}X_2 + \cdots + a_{pp}X_p
\end{aligned}$$

It can be shown that

$$Var(Y_i) = \mathbf{a}_i'\mathbf{\Sigma}\mathbf{a}_i, \;\; i = 1, 2, \cdots, p$$

and

$$Cov(Y_i, Y_k) = \mathbf{a}_i'\mathbf{\Sigma}\mathbf{a}_k, \;\; i, k = 1, 2, \cdots, p$$

The principal components are those uncorrelated linear combinations whose variances are as large as possible.

The first principal component is the linear combination with maximum variance.That is, it maximizes $Var(Y_1) = \mathbf{a}_1'\mathbf{\Sigma}\mathbf{a}_1$. It is clear that $Var(Y_1) = \mathbf{a}_1'\mathbf{\Sigma}\mathbf{a}_1$ can be increased by multiplying any by some constant.To eliminate this indeterminacy, it is convenient to restrict attention to coefficient vectors of unit length.We therefore define

- First principal component = linear combination $\mathbf{a}_1'\mathbf{X}$ that maximizes $Var(\mathbf{a}_1'\mathbf{X})$ subject to $\mathbf{a}_1'\mathbf{a}_1 = 1$.
- Second principal component = linear combination $\mathbf{a}_2'\mathbf{X}$ that maximizes $Var(\mathbf{a}_2'\mathbf{X})$ subject to $\mathbf{a}_2'\mathbf{a}_2 = 1$ and $Cov(\mathbf{a}_1'\mathbf{X}, \mathbf{a}_2'\mathbf{X}) = 0$.

At the i^{th} step,

- i^{th} principal component = linear combination $\mathbf{a}_i'\mathbf{X}$ that maximizes $Var(\mathbf{a}_i'\mathbf{X})$ subject to $\mathbf{a}_i'\mathbf{a}_i = 1$ and $Cov(\mathbf{a}_i'\mathbf{X}, \mathbf{a}_k'\mathbf{X}) = 0$, for $k < i$.

Remarks

1. Let $\mathbf{\Sigma}$ be the covariance matrix associated with the random vector $\mathbf{X}' = [X_1, X_2, \cdots, X_p]$. Let $\mathbf{\Sigma}$ have the eigenvalue-eigenvector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \cdots, (\lambda_p, \mathbf{e}_p)$ where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$. Then the i^{th} principal component is given by

$$Y_i = \mathbf{e}_i'\mathbf{X} = e_{i1}X_1 + e_{i2}X_2 + \cdots + e_{ip}X_p, \ \ i = 1, 2, \cdots, p$$

With these choices,

$$Var(Y_i) = \mathbf{e}_i'\mathbf{\Sigma}\mathbf{e}_i, \ \ i = 1, 2, \cdots, p$$

and

$$Cov(Y_i, Y_k) = \mathbf{e}_i'\mathbf{\Sigma}\mathbf{e}_k = 0, \ \ i \neq k$$

If some λ_i are equal, the choices of the corresponding coefficient vectors, $\mathbf{e}_i$, and hence Y_i are not unique.

This result implies that the principal components are uncorrelated and have variances equal to the eigenvalues of $\mathbf{\Sigma}$.

2. Let $\mathbf{X}' = [X_1, X_2, \cdots, X_p]$ have covariance matrix $\mathbf{\Sigma}$ eigenvalue-eigenvector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \cdots, (\lambda_p, \mathbf{e}_p)$ where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$. Let $Y_1 = \mathbf{e}_1'\mathbf{X}, Y_2 = \mathbf{e}_2'\mathbf{X}, \cdots, Y_p = \mathbf{e}_p'\mathbf{X}$ be the principal components. Then

$$\sigma_{11} + \sigma_{22} + \cdots + \sigma_{pp} = \sum_{i=1}^{p} Var(X_i) = \lambda_1 + \lambda_2 + \cdots + \lambda_p = \sum_{i=1}^{p} Var(Y_i)$$

This results says that

$$\begin{aligned} \text{Total population variance} &= \sigma_{11} + \sigma_{22} + \cdots + \sigma_{pp} \\ &= \lambda_1 + \lambda_2 + \cdots + \lambda_p \end{aligned}$$

and consequently, the proportion of total variance due to (explained by) the k^{th} principal component is given by

$$\frac{\lambda_k}{\lambda_1 + \lambda_2 + \cdots + \lambda_p}, \ \ k = 1, 2, \cdots, p$$

If most (for instance, 80 to 90%) of the total population variance, for large p, can be attributed to the first one, two, or three components, then these components can **replace** the original p variables without much loss of information.

Each component of the coefficient vector $\mathbf{e}_i' = [e_{i1}, e_{i2}, \cdots, e_{ip}]$ also merits inspection.The magnitude of e_{ik} measures the importance of the k^{th} variable to the i^{th} principal component, irrespective of the other variables. In particular, e_{ik} is proportional to the correlation coefficient between Y_i and X_k.

3. If $Y_1 = \mathbf{e}_1'\mathbf{X}, Y_2 = \mathbf{e}_2'\mathbf{X}, \cdots, Y_p = \mathbf{e}_p'\mathbf{X}$ are the principal components obtained from the covariance matrix $\mathbf{\Sigma}$, then

$$\rho_{Y_i,X_k} = \frac{e_{ik}\sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}, \quad i,k = 1,2,\cdots,p$$

are the correlation coefficients between the components Y_i and the variables X_k. Here $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \cdots, (\lambda_p, \mathbf{e}_p)$ are the eigenvalue–eigenvector pairs for $\mathbf{\Sigma}$.

Although the correlations of the variables with the principal components often help to interpret the components, they measure only the univariate contribution of an individual X to a component Y. That is, they do not indicate the importance of an X to a component Y in the presence of the other X's. For this reason, some statisticians recommend that only the coefficients e_{ik} and not the correlations, be used to interpret the components. In practice, variables with relatively large coefficients (in absolute value) tend to have relatively large correlations, so the two measures of importance, the first multivariate and the second univariate, frequently give similar results. Therefore, it is suggested that both the coefficients and the correlations be examined to help interpret the principal components.

Example 3.1.1

Suppose the random variables X_1, X_2, and X_3 and have the covariance matrix

$$\mathbf{\Sigma} = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

It can be verified using the following R codes that the eigenvalues and the associated eigenvectors of $\mathbf{\Sigma}$ are, respectively:

```
Sigma <- matrix(c(1,-2,0,-2,5,0,0,0,2),nrow = 3, byrow=TRUE)
eigen(Sigma) #generates the eigenvalues and eigenvectors
```

$$\lambda_1 = 5.83,\ \lambda_2 = 2.00,\ \lambda_3 = 0.17$$

and

$$\begin{aligned}
\mathbf{e}_1' &= [-0.383, 0.924, 0] \\
\mathbf{e}_2' &= [0, 0, 1] \\
\mathbf{e}_1' &= [0.924, 0.383, 0]
\end{aligned}$$

Therefore, the principal components become

$$\begin{aligned}
Y_1 &= -0.383X_1 + 0.924X_2 \\
Y_2 &= X_3 \\
Y_3 &= 0.924X_1 + 0.383X_2
\end{aligned}$$

Now,

$$\begin{aligned}
Var(Y_1) &= Var(-0.383X_1 + 0.924X_2) \\
&= (-0.383)^2 Var(X_1) + (0.924)^2 Var(X_2) + 2(-0.383)(0.924)Cov(X_1, X_2) \\
&= (-0.383)^2(1) + (0.924)^2(5) + 2(-0.383)(0.924)(-2) \\
&\approx 5.83 \\
&= \lambda_1, \quad \text{and} \\
Cov(Y_1, Y_2) &= Cov(-0.383X_1 + 0.924X_2, X_3) \\
&= -0.383Cov(X_1, X_3) + 0.942Cov(X_2, X_3) \\
&= -0.383(0) + 0.942(0) \\
&= 0
\end{aligned}$$

It is also readily apparent that

$$\sigma_{11} + \sigma_{22} + \sigma_{33} = 1 + 5 + 2 = \lambda_1 + \lambda_2 + \lambda_3 = 5.83 + 2.00 + 0.17 = 8.$$

Thus, the proportion of the total variance accounted for by the first principal component is

$$\frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} = \frac{5.83}{8} = 0.7288.$$

Further, the first two components account for

$$\frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \lambda_3} = \frac{5.83 + 2.00}{8} = 0.9788.$$

of the population variance. This implies that the first two components could replace the original three variables with little loss of information.

Next,

$$\rho_{Y_1,X_1} = \frac{e_{11}\sqrt{\lambda_1}}{\sqrt{\sigma_{11}}} = \frac{-0.383\sqrt{5.83}}{\sqrt{1}} \approx -0.9248$$

and

$$\rho_{Y_1,X_2} = \frac{e_{12}\sqrt{\lambda_1}}{\sqrt{\sigma_{22}}} = \frac{0.924\sqrt{5.83}}{\sqrt{5}} \approx 0.9977$$

Notice here that X_2 with coefficient 0.924 receives the greatest weight in the component Y_1. It also has the largest correlation (in absolute value) with Y_1. The correlation of X_1 with Y_1 is almost as large as that for X_2 indicating that the variables are about equally important to the first principal component. The relative sizes of the coefficients of X_1 and X_2 suggest, however, that X_2 contributes more to the determination of Y_1 than X_1 does. Since, in this case, both coefficients are reasonably large and they have opposite signs, we would argue that both variables aid in the interpretation of Y_1.

Finally,

$$\rho_{Y_2,X_1} = \rho_{Y_2,X_2} = 0, \text{ and } \rho_{Y_2,X_3} = \frac{e_{23}\sqrt{\lambda_2}}{\sqrt{\sigma_{22}}} = \frac{1(\sqrt{2})}{\sqrt{2}} = 1.$$

The remaining correlations can be neglected, since the third component is unimportant.

Principal Components Obtained from Standardized Variables

Principal components may also be obtained for the standardized variables

$$\begin{aligned} Z_1 &= \frac{X_1 - \mu_1}{\sqrt{\sigma_{11}}} \\ Z_2 &= \frac{X_2 - \mu_2}{\sqrt{\sigma_{22}}} \\ Z_3 &= \frac{X_3 - \mu_3}{\sqrt{\sigma_{33}}} \\ &\vdots \\ Z_p &= \frac{X_p - \mu_p}{\sqrt{\sigma_{pp}}} \end{aligned}$$

In matrix notation,

$$\mathbf{Z} = (\mathbf{V}^{1/2})^{-1}(\mathbf{X} - \boldsymbol{\mu})$$

where $\mathbf{V}^{1/2}$ is the $p \times p$ standard deviation matrix

$$\mathbf{V}^{1/2} = \begin{bmatrix} \sqrt{\sigma_{11}} & 0 & \cdots & 0 \\ 0 & \sqrt{\sigma_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{\sigma_{pp}} \end{bmatrix}$$

Clearly,

$$E(\mathbf{Z}) = \mathbf{0}$$

and

$$Cov(\mathbf{Z}) = (\mathbf{V}^{1/2})^{-1}\mathbf{\Sigma}(\mathbf{V}^{1/2})^{-1} = \boldsymbol{\rho}$$

The principal components of $\mathbf{Z}$ may be obtained from the eigenvectors of the correlation matrix $\boldsymbol{\rho}$ of $\mathbf{X}$.

All our previous results apply, with some simplifications, since the variance of each Z_i is unity. We shall continue to use the notation Y_i to refer to the i^{th} principal component and $(\lambda_i, \mathbf{e}_i)$ for the eigenvalue–eigenvector pair from either $\boldsymbol{\rho}$ or $\mathbf{\Sigma}$. However, the $(\lambda_1, \mathbf{e}_i)$ derived from $\mathbf{\Sigma}$ are, in general, not the same as the ones derived from $\boldsymbol{\rho}$.

Remark

The i^{th} principal component of the standardized variables $\mathbf{Z}' = [Z_1, Z_2, \cdots, Z_p]$ with $Cov(\mathbf{Z}) = \boldsymbol{\rho}$ is given by

$$Y_i = \mathbf{e_i}'\mathbf{Z} = \mathbf{e}_i'(\mathbf{V}^{1/2})^{-1}(\mathbf{X} - \boldsymbol{\mu}), \ i = 1, 2, \cdots, p$$

Moreover,

$$\sum_{i=1}^{p} Var(Y_i) = \sum_{i=1}^{p} Var(Z_i) = p$$

and

$$\rho_{Y_i, Z_k} = e_{ik}\sqrt{\lambda_i}, \ i, k = 1, 2, \cdots, p$$

In this case, $(\lambda_i, \mathbf{e}_i)$ are the eigenvalue–eigenvector pair for $\boldsymbol{\rho}$ with

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0.$$

Since the total (standardized variables) population variance is simply p, the sum of the diagonal elements of the correlation matrix $\boldsymbol{\rho}$. Thus, the proportion of total variance explained by the k^{th} principal component of $\mathbf{Z}$ is $\frac{\lambda_k}{p}, k = 1, 2, \cdots, p$, where λ_k are the eigenvalues of $\boldsymbol{\rho}$.

Example 3.1.2

Consider the covariance matrix

$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 4 \\ 4 & 100 \end{bmatrix}$$

and the derived correlation matrix

$$\boldsymbol{\rho} = \begin{bmatrix} 1 & 0.4 \\ 0.4 & 1 \end{bmatrix}$$

The eigenvalue–eigenvector pairs from $\boldsymbol{\Sigma}$ are

$$\lambda_1 = 100.16; \ \ \lambda_2 = 0.84$$

and

$$\begin{aligned} \mathbf{e}_1' &= [0.040, 0.999] \\ \mathbf{e}_2' &= [-0.999, 0.040] \end{aligned}$$

Therefore, the respective principal components become

$$\begin{aligned} Y_1 &= 0.040X_1 + 0.999X_2 \\ Y_2 &= -0.999X_1 + 0.04X_2 \end{aligned}$$

Similarly, the eigenvalue–eigenvector pairs from $\boldsymbol{\rho}$ are

$$\lambda_1 = 1.4; \ \ \lambda_2 = 0.6$$

and

$$\begin{aligned} \mathbf{e}_1' &= [0.707, 0.707] \\ \mathbf{e}_2' &= [-0.707, 0.707] \end{aligned}$$

and

$$\begin{aligned} Y_1 &= 0.707Z_1 + 0.707Z_2 = 0.707(X_1 - \mu_1) + 0.0707(X_2 - \mu_2) \\ Y_2 &= -0.707Z_1 + 0.707Z_2 = -0.707(X_1 - \mu_1) + 0.0707(X_2 - \mu_2) \end{aligned}$$

Notice that because of its large variance, completely dominates the first principal component determined from $\boldsymbol{\Sigma}$. In fact, the first principal component explains a proportion

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{100.16}{100.16 + 0.84} = 0.992$$

However, When the variables X_1 and X_2 are standardized, the resulting variables contribute equally to the principal components determined from $\boldsymbol{\rho}$. For example,

$$\rho_{Y_1,Z_1} = e_{11}\sqrt{\lambda_1} = 0.707\sqrt{1.4} = 0.837, \text{ and} \rho_{Y_1,Z_2} = e_{12}\sqrt{\lambda_1} = 0.707\sqrt{1.4} = 0.837$$

In this case, the first principal component explains only a proportion

$$\frac{\lambda_1}{p} = \frac{1.4}{2} = 0.70$$

of the total (standardized) population variance.

Most strikingly, we see that the relative importance of the variables to, for instance, the first principal component is greatly affected by the standardization. When the first principal component obtained from $\boldsymbol{\rho}$ is expressed in terms of X_1 and X_2 the relative magnitudes of the weights 0.707 and 0.0707 are in direct opposition to those of the weights 0.040 and 0.999 attached to these variables in the principal component obtained from $\boldsymbol{\Sigma}$.

This example demonstrates that the principal components derived from $\boldsymbol{\Sigma}$ are different from those derived from $\boldsymbol{\rho}$. Furthermore, one set of principal components is not a simple function of the other. This suggests that the standardization is not inconsequential.

Variables should probably be standardized if they are measured on scales with widely differing ranges or if the units of measurement are not commensurate. For example, if X_1 represents annual sales in the PhP10,000 to PhP350,000 range and X_2 is the ratio (net annual income)/(total assets) that falls in the 0.01 to 0.60 range, then the total variation will be due almost exclusively to sales. In this case, we would expect a single (important) principal component with a heavy weighting of X_1. Alternatively, if both variables are standardized, their subsequent magnitudes will be of the same order, and X_2 (or Z_2) will play a larger role in the construction of the principal components.

Summarizing Sample Variation by Principal Components

Suppose the data $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n$ represent n independent drawings from some p-dimensional population with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. These data yield the sample mean vector $\overline{\mathbf{x}}$ the sample covariance matrix $\mathbf{S}$, and the sample correlation matrix $\mathbf{R}$.

Our objective in this section will be to construct uncorrelated linear combinations of the measured characteristics that account for much of the variation in the sample.The uncorrelated combinations with the largest variances will be called the *sample principal components.*

The sample principal components are defined as those linear combinations which have maximum sample variance. As with the population quantities, we restrict the coefficient vectors $\mathbf{a}_i$ to satisfy $\mathbf{a}_i'\mathbf{a}_i = 1$. Specifically

- First sample principal component = linear combination $\mathbf{a}_1'\mathbf{x}_j$ that maximizes the sample variance of $\mathbf{a}_i'\mathbf{x}_j$ subject to $\mathbf{a}_1'\mathbf{a}_1 = 1$.

- Second sample principal component = linear combination $\mathbf{a}_2'\mathbf{x}_j$ that maximizes the sample variance of $\mathbf{a}_2'\mathbf{x}_j$ subject to $\mathbf{a}_2'\mathbf{a}_2 = 1$ and zero sample covariance for the pairs $(\mathbf{a}_1'\mathbf{x}_j, \mathbf{a}_2'\mathbf{x}_j)$.

At the i^{th} step, we have

- i^{th} sample principal component = linear combination $\mathbf{a}_i'\mathbf{x}_j$ that maximizes the sample variance of $\mathbf{a}_i'\mathbf{x}_j$ subject to $\mathbf{a}_i'\mathbf{a}_i = 1$ and zero sample covariance for the pairs $(\mathbf{a}_i'\mathbf{x}_j, \mathbf{a}_k'\mathbf{x}_j), k < i$.

In summary, if $\mathbf{S} = \{s_{ik}\}$ is the $p \times p$ sample covariance matrix with eigenvalue-eigenvector pairs $(\hat{\lambda}_i, \hat{\mathbf{e}}_i),\ i = 1, 2, \cdots, p$,then the i^{th} sample principal components is given by

$$\hat{y}_i = \hat{\mathbf{e}}_i\mathbf{x} = \hat{e}_{i1}x_1 + \hat{e}_{i2}x_2 + \cdots + \hat{e}_{ip}x_p,\ i = 1, 2, \cdots, p$$

where $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p$ and $\mathbf{x}$ is s any observation on the variables $X_1, X_2, \cdots, X_p$. Also,

$$\begin{aligned} \text{Sample variance}(\hat{y}_i) &= \hat{\lambda}_i,\ i = 1, 2, \cdots, p \\ \text{Sample covariance}(\hat{y}_i, \hat{y}_k) &= 0,\ i \neq k \end{aligned}$$

In addition,

$$\text{Total sample variance} = \sum_{i=1}^{p} s_{ii} = \sum_{i=1}^{p} \hat{\lambda}_i$$

and

$$r_{(\hat{y}_i, x_k)} = \frac{\hat{e}_{ik}\sqrt{\hat{\lambda}_i}}{\sqrt{s_{kk}}},\ i, k = 1, 2, \cdots, p$$

Example 3.1.3

A census provided information, by barangay, on five socioeconomic variables: X_1 = total population (thousands), X_2 = professional degree (%), X_3 = employed age over 16 (%), X_4 = government employment (%), and X_5 = median home value (PhP100,000). Summary statistics for 61 barangays yielded the following:

$$\overline{\mathbf{x}}' = [4.47, 3.96, 71.42, 26.91, 1.64]$$

and

$$\mathbf{S} = \begin{bmatrix} 3.397 & -1.102 & 4.306 & -2.078 & 0.027 \\ -1.102 & 9.673 & -1.513 & 10.953 & 1.203 \\ 4.306 & -1.513 & 55.626 & -28.937 & -0.044 \\ -2.078 & 10.953 & -28.937 & 89.067 & 0.957 \\ 0.027 & 1.203 & -0.044 & 0.957 & 0.319 \end{bmatrix}$$

Using similar code chunk we obtain the eigenvalues and eigenvectors of **S** and these are given below.

```
Sigma <- matrix(c(3.397, -1.102, 4.306, -2.078, 0.027,
                  -1.102,  9.673, -1.513, 10.953, 1.203,
                  4.306, -1.513, 55.626, -28.937, -0.044,
                  -2.078, 10.953, -28.937, 89.067, 0.957,
                  0.027, 1.203, -0.044, 0.957, 0.319),
                nrow = 5,
                byrow=TRUE)
eigen(Sigma) #generates the eigenvalues and eigenvectors
```

$$\hat{\lambda}_1 = 107.02,\ \hat{\lambda}_2 = 39.67,\ \hat{\lambda}_3 = 8.37,\ \hat{\lambda}_4 = 2.87, \hat{\lambda}_5 = 0.16$$

and

$$\begin{aligned}
\hat{\mathbf{e}}_1' &= [0.039, -0.105, 0.492, -0.863, -0.009] \\
\hat{\mathbf{e}}_2' &= [-0.071, -0.130, -0.864, -0.480, -0.015] \\
\hat{\mathbf{e}}_3' &= [-0.188, 0.961, -0.046, -0.153, 0.125] \\
\hat{\mathbf{e}}_4' &= [0.977, 0.171, -0.091, -0.30, 0.082] \\
\hat{\mathbf{e}}_5' &= [-0.058, -0.139, 0.005, 0.007, 0.989]
\end{aligned}$$

Therefore, the sample principal components are:

$$\begin{aligned}
\hat{y}_1 &= 0.039x_1 - 0.105x_2 + 0.492x_3 - 0.863x_4 - 0.009x_5 \\
\hat{y}_2 &= -0.071x_1 - 0.130x_2 - 0.864x_3 - 0.480x_4 - 0.015x_5 \\
\hat{y}_3 &= -0.188x_2 + 0.961x_2 - 0.046x_3 - 0.153x_4 + 0.125x_5 \\
\hat{y}_4 &= 0.977x_1 + 0.171x_2 - 0.091x_3 - 0.300x_4 + 0.082x_5 \\
\hat{y}_5 &= -0.058x_1 - 0.139x_2 + 0.005x_3 + 0.007x_4 + 0.989x_5
\end{aligned}$$

The proportion of the total sample variance accounted for by the first two sample principal components are, respectively:

$$\frac{\hat{\lambda}_1}{\hat{\lambda}_1 + \hat{\lambda}_2 + \hat{\lambda}_3 + \hat{\lambda}_4 + \hat{\lambda}_5} = \frac{107.02}{107.02 + 39.67 + 8.37 + 2.87 + 0.16} \approx 0.6769,$$

and

$$\frac{\hat{\lambda}_2}{\hat{\lambda}_1 + \hat{\lambda}_2 + \hat{\lambda}_3 + \hat{\lambda}_4 + \hat{\lambda}_5} = \frac{39.67}{107.02 + 39.67 + 8.37 + 2.87 + 0.16} \approx 0.2509,$$

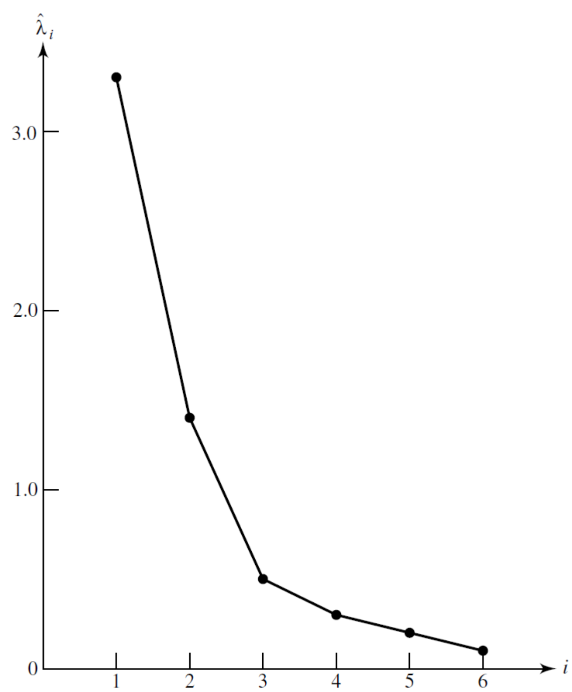
The first principal component explains 67.69% of the total sample variance. The first two principal components, collectively, explain 92.78% of the total sample variance. Consequently, sample variation is summarized very well by two principal components and a reduction in the data from 61 observations on 5 observations to 61 observations on 2 principal components is reasonable.

Given the foregoing component coefficients, the first principal component, appears to be essentially a weighted difference between the percent employed by government and the percent total employment. The second principal component appears to be a weighted sum of the two.

The Number of Principal Components

There is always the question of how many components to retain.There is no definitive answer to this question.Things to consider include the amount of total sample variance explained, the relative sizes of the eigenvalues (the variances of the sample components), and the subject-matter interpretations of the components.

A useful visual aid to determining an appropriate number of principal components is a scree plot.3With the eigenvalues ordered from largest to smallest, a scree plot is a plot of versus i—the magnitude of an eigenvalue versus its number. To determine the appropriate number of components, we look for an elbow (bend) in the scree plot. The number of components is taken to be the point at which the remaining eigenvalues are relatively small and all about the same size. The figure below shows a scree plot for a situation with six principal components.



An elbow occurs in the plot in the figure at about $i = 3$. That is, the eigenvalues after $\hat{\lambda}_2$ are all relatively small and about the same size. In this case, it appears, without any other evidence, that two (or perhaps three) sample principal components effectively summarize the total sample variance.

Alternatively, Horn's Parallel Analysis is used to determine how many principal components to keep in an analysis. It works by comparing the eigenvalues from your actual data to those from randomly generated data of the same size, retaining components whose eigenvalues from the real data are greater than those from the random data.

Performing PCA in R

There are many functions in R for performing PCA. Some of these include *prcomp()*, *princomp()*, and *PCA()* in **FactoMiner** package. No matter what function you decide to use, you can easily extract and visualize the results of PCA using R functions provided in the **factoextra** package.

The primary difference between these functions lies on the method used in the PCA extraction. The function *princomp()* uses the spectral decomposition approach. The functions *prcomp()* and *PCA()* use the singular value decomposition (SVD). Spectral decomposition which examines the covariances/correlations between variables while singular value decomposition which examines the covariances/correlations between individuals.

Arguments for *prcomp()*:

- *x*: a numeric matrix or data frame
- *scale*: a logical value indicating whether the variables should be scaled to have unit variance before the analysis takes place

Arguments for *princomp()*:

- *x*: a numeric matrix or data frame
- *cor*: a logical value. If TRUE, the data will be centered and scaled before the analysis
- *scores*: a logical value. If TRUE, the coordinates on each principal component are calculated

The elements of the outputs returned by the functions prcomp() and princomp() includes:

prcomp() name	princomp() name	Description
sdev	sdev	the standard deviations of the principal components
rotation	loadings	the matrix of variable loadings (columns are eigenvectors)
center	center	the variable means (means that were substracted)
scale	scale	the variable standard deviations (the scaling applied to each variable)
x	scores	The coordinates of the individuals (observations) on the principal components.

For this quick demo of PCA in R, we shall use the demo data sets *decathlon2* from the **factoextra** package. Briefly, it contains:

- Active individuals (rows 1 to 23) and active variables (columns 1 to 10), which are used to perform the principal component analysis
- Supplementary individuals (rows 24 to 27) and supplementary variables (columns 11 to 13), which coordinates will be predicted using the PCA information and parameters obtained with active individuals/variables.

```r
library(FactoMineR)
library(factoextra)
data(decathlon2)
decathlon2.active <- decathlon2[1:23, 1:10]
head(decathlon2.active[, 1:6])
```

```
##           X100m Long.jump Shot.put High.jump X400m X110m.hurdle
## SEBRLE    11.04      7.58    14.83      2.07 49.81        14.69
## CLAY      10.76      7.40    14.26      1.86 49.37        14.05
## BERNARD   11.02      7.23    14.25      1.92 48.93        14.99
## YURKOV    11.34      7.09    15.19      2.10 50.42        15.31
## ZSIVOCZKY 11.13      7.30    13.48      2.01 48.62        14.17
## McMULLEN  10.83      7.31    13.76      2.13 49.91        14.38
```

Compute PCA in R using prcomp()

Let us begin using the *prcomp()* function and the **factoextra** package.

```
#Extract the PCs
res.pca <- prcomp(decathlon2.active, scale = TRUE)
```

Access to the PCA results

```
# Eigenvalues
(eig.val <- get_eigenvalue(res.pca))
```

```
##         eigenvalue variance.percent cumulative.variance.percent
## Dim.1    4.1242133        41.242133                    41.24213
## Dim.2    1.8385309        18.385309                    59.62744
## Dim.3    1.2391403        12.391403                    72.01885
## Dim.4    0.8194402         8.194402                    80.21325
## Dim.5    0.7015528         7.015528                    87.22878
## Dim.6    0.4228828         4.228828                    91.45760
## Dim.7    0.3025817         3.025817                    94.48342
## Dim.8    0.2744700         2.744700                    97.22812
## Dim.9    0.1552169         1.552169                    98.78029
## Dim.10   0.1219710         1.219710                   100.00000
```

```
# Eigenvectors
(eigen.vec <- res.pca$rotation)
```

```
##                      PC1         PC2         PC3         PC4        PC5
## X100m       -0.418859080 -0.13230683 -0.27089959  0.03708806 -0.2321476
## Long.jump    0.391064807  0.20713320  0.17117519 -0.12746997  0.2783669
## Shot.put     0.361388111  0.06298590 -0.46497777  0.14191803 -0.2970589
## High.jump    0.300413236 -0.34309742 -0.29652805  0.15968342  0.4807859
## X400m       -0.345478567  0.21400770 -0.25470839  0.47592968  0.1240569
## X110m.hurdle -0.376265119 -0.01824645 -0.40325254 -0.01866477  0.2676975
## Discus       0.365965721  0.03662510 -0.15857927  0.43636361 -0.4873988
## Pole.vault  -0.106985591  0.59549862 -0.08449563 -0.37447391 -0.2646712
## Javeline     0.210864329  0.28475723 -0.54270782 -0.36646463  0.2361698
## X1500m       0.002106782  0.57855748  0.19715884  0.49491281  0.3142987
##                      PC6         PC7         PC8         PC9        PC10
## X100m       -0.054398099 -0.16604375 -0.19988005 -0.76924639  0.12718339
## Long.jump    0.051865558 -0.28056361 -0.75850657 -0.13094589  0.08509665
## Shot.put     0.368739186 -0.01797323  0.04649571  0.12129309  0.62263702
```

```
## High.jump     0.437716883  0.05118848  0.16111045 -0.28463225 -0.38244596
## X400m         0.075796432  0.52012255 -0.44579641  0.20854176 -0.09784197
## X110m.hurdle -0.004048005 -0.67276768 -0.01592804  0.41058421 -0.04475363
## Discus       -0.305315353 -0.25946615 -0.07550934  0.03391600 -0.49418361
## Pole.vault    0.503563524 -0.01889413  0.06282691 -0.06540692 -0.39288155
## Javeline     -0.556821016  0.24281145  0.10086127 -0.10268134 -0.01103627
## X1500m       -0.064663250 -0.20245828  0.37119711 -0.25950868  0.17991689
```

```r
# Results for Variables
res.var <- get_pca_var(res.pca)
res.var$coord          # Coordinates
```

```
##                     Dim.1       Dim.2       Dim.3       Dim.4      Dim.5
## X100m        -0.850625692 -0.17939806 -0.30155643  0.03357320 -0.1944440
## Long.jump     0.794180641  0.28085695  0.19054653 -0.11538956  0.2331567
## Shot.put      0.733912733  0.08540412 -0.51759781  0.12846837 -0.2488129
## High.jump     0.610083985 -0.46521415 -0.33008517  0.14455012  0.4027002
## X400m        -0.701603377  0.29017826 -0.28353292  0.43082552  0.1039085
## X110m.hurdle -0.764125197 -0.02474081 -0.44888733 -0.01689589  0.2242200
## Discus        0.743209016  0.04966086 -0.17652518  0.39500915 -0.4082391
## Pole.vault   -0.217268042  0.80745110 -0.09405773 -0.33898477 -0.2216853
## Javeline      0.428226639  0.38610928 -0.60412432 -0.33173454  0.1978128
## X1500m        0.004278487  0.78448019  0.21947068  0.44800961  0.2632527
##                     Dim.6        Dim.7        Dim.8       Dim.9       Dim.10
## X100m        -0.035374780 -0.091336386 -0.104716925 -0.30306448  0.044417974
## Long.jump     0.033727883 -0.154330810 -0.397380703 -0.05158951  0.029719453
## Shot.put      0.239789034 -0.009886612  0.024359049  0.04778655  0.217451948
## High.jump     0.284644846  0.028157465  0.084405578 -0.11213822 -0.133566774
## X400m         0.049289996  0.286106008 -0.233552216  0.08216041 -0.034170673
## X110m.hurdle -0.002632395 -0.370072158 -0.008344682  0.16176025 -0.015629914
## Discus       -0.198544870 -0.142725641 -0.039559255  0.01336209 -0.172590426
## Pole.vault    0.327464549 -0.010393176  0.032914942 -0.02576874 -0.137211339
## Javeline     -0.362097598  0.133564318  0.052841099 -0.04045397 -0.003854347
## X1500m       -0.042050151 -0.111367083  0.194469730 -0.10224014  0.062834809
```

```r
res.var$contrib        # Contributions to the PCs
```

```
##                     Dim.1      Dim.2      Dim.3       Dim.4     Dim.5
## X100m        1.754429e+01  1.7505098  7.3386590  0.13755240  5.389252
## Long.jump    1.529317e+01  4.2904162  2.9300944  1.62485936  7.748815
## Shot.put     1.306014e+01  0.3967224 21.6204325  2.01407269  8.824401
## High.jump    9.024811e+00 11.7715838  8.7928883  2.54987951 23.115504
## X400m        1.193554e+01  4.5799296  6.4876363 22.65090599  1.539012
## X110m.hurdle 1.415754e+01  0.0332933 16.2612611  0.03483735  7.166193
```

```
## Discus       1.339309e+01  0.1341398  2.5147385 19.04132022 23.755756
## Pole.vault   1.144592e+00 35.4618611  0.7139512 14.02307063  7.005084
## Javeline     4.446377e+00  8.1086683 29.4531777 13.42963254  5.577615
## X1500m       4.438531e-04 33.4728757  3.8871610 24.49386930  9.878367
##                    Dim.6       Dim.7       Dim.8      Dim.9      Dim.10
## X100m         0.295915322  2.75705260  3.99520353 59.1740009  1.61756139
## Long.jump     0.269003613  7.87159392 57.53322220  1.7146826  0.72414393
## Shot.put     13.596858744  0.03230371  0.21618512  1.4712015 38.76768578
## High.jump    19.159607001  0.26202607  2.59565787  8.1015517 14.62649091
## X400m         0.574509906 27.05274658 19.87344405  4.3489667  0.95730504
## X110m.hurdle  0.001638634 45.26163460  0.02537025 16.8579392  0.20028870
## Discus        9.321746508  6.73226823  0.57016606  0.1150295 24.42174410
## Pole.vault   25.357622290  0.03569883  0.39472201  0.4278065 15.43559151
## Javeline     31.004964393  5.89573984  1.01729950  1.0543458  0.01217993
## X1500m        0.418133591  4.09893563 13.77872941  6.7344755  3.23700871
```

```r
res.var$cos2           # Quality of representation
```

```
##                     Dim.1        Dim.2       Dim.3        Dim.4      Dim.5
## X100m        7.235641e-01 0.0321836641 0.090936280 0.0011271597 0.03780845
## Long.jump    6.307229e-01 0.0788806285 0.036307981 0.0133147506 0.05436203
## Shot.put     5.386279e-01 0.0072938636 0.267907488 0.0165041211 0.06190783
## High.jump    3.722025e-01 0.2164242070 0.108956221 0.0208947375 0.16216747
## X400m        4.922473e-01 0.0842034209 0.080390914 0.1856106269 0.01079698
## X110m.hurdle 5.838873e-01 0.0006121077 0.201499837 0.0002854712 0.05027463
## Discus       5.523596e-01 0.0024662013 0.031161138 0.1560322304 0.16665918
## Pole.vault   4.720540e-02 0.6519772763 0.008846856 0.1149106765 0.04914437
## Javeline     1.833781e-01 0.1490803723 0.364966189 0.1100478063 0.03912992
## X1500m       1.830545e-05 0.6154091638 0.048167378 0.2007126089 0.06930197
##                     Dim.6        Dim.7        Dim.8        Dim.9       Dim.10
## X100m        1.251375e-03 0.0083423353 1.096563e-02 0.0918480768 1.972956e-03
## Long.jump    1.137570e-03 0.0238179990 1.579114e-01 0.0026614779 8.832459e-04
## Shot.put     5.749878e-02 0.0000977451 5.933633e-04 0.0022835540 4.728535e-02
## High.jump    8.102269e-02 0.0007928428 7.124302e-03 0.0125749811 1.784008e-02
## X400m        2.429504e-03 0.0818566479 5.454664e-02 0.0067503333 1.167635e-03
## X110m.hurdle 6.929502e-06 0.1369534023 6.963371e-05 0.0261663784 2.442942e-04
## Discus       3.942007e-02 0.0203706085 1.564935e-03 0.0001785453 2.978746e-02
## Pole.vault   1.072330e-01 0.0001080181 1.083393e-03 0.0006640282 1.882695e-02
## Javeline     1.311147e-01 0.0178394271 2.792182e-03 0.0016365234 1.485599e-05
## X1500m       1.768215e-03 0.0124026272 3.781848e-02 0.0104530472 3.948213e-03
```

```r
# Results for individuals
res.ind <- get_pca_ind(res.pca)
res.ind$coord          # Coordinates
```

```
##                  Dim.1      Dim.2       Dim.3       Dim.4         Dim.5
## SEBRLE       0.1912074  1.5541282 -0.62836882  0.08205241  1.1426139415
## CLAY         0.7901217  2.4204156  1.35688701  1.26984296 -0.8068483724
## BERNARD     -1.3292592  1.6118687 -0.19614996 -1.92092203  0.0823428202
## YURKOV      -0.8694134 -0.4328779 -2.47398223  0.69723814  0.3988584116
## ZSIVOCZKY   -0.1057450 -2.0233632  1.30493117 -0.09929630 -0.1970241089
## McMULLEN     0.1185550 -0.9916237  0.84355824  1.31215266  1.5858708644
## MARTINEAU   -2.3923532 -1.2849234 -0.89816842  0.37309771 -2.2433515889
## HERNU       -1.8910497  1.1784614 -0.15641037  0.89130068 -0.1267412520
## BARRAS      -1.7744575 -0.4125321  0.65817750  0.22872866 -0.2338366980
## NOOL        -2.7770058 -1.5726757  0.60724821 -1.55548081  1.4241839810
## BOURGUIGNON -4.4137335  1.2635770 -0.01003734  0.66675478  0.4191518468
## Sebrle       3.4514485  1.2169193 -1.67816711 -0.80870696 -0.0250530746
## Clay         3.3162243  1.6232908 -0.61840443 -0.31679906  0.5691645854
## Karpov       4.0703560 -0.7983510  1.01501662  0.31336354 -0.7974259553
## Macey        1.8484623 -2.0638828 -0.97928455  0.58469073 -0.0002157834
## Warners      1.3873514  0.2819083  1.99969621 -1.01959817 -0.0405401497
## Zsivoczky    0.4715533 -0.9267436 -1.72815525 -0.18483138  0.4073029909
## Hernu        0.2763118 -1.1657260  0.17056375 -0.84869401 -0.6894795441
## Bernard      1.3672590 -1.4780354  0.83137913  0.74531557  0.8598016482
## Schwarzl    -0.7102777  0.6584251  1.04075176 -0.92717510 -0.2887568007
## Pogorelov   -0.2143524  0.8610557  0.29761010  1.35560294 -0.0150531057
## Schoenbeck  -0.4953166  1.3000530  0.10300360 -0.24927712 -0.6452257128
## Barras      -0.3158867 -0.8193681 -0.86169481 -0.58935985 -0.7797389436
##                   Dim.6       Dim.7        Dim.8        Dim.9      Dim.10
## SEBRLE       0.46389755 -0.20796012  0.043460568 -0.659344137  0.03273238
## CLAY        -1.30420016 -0.21291866  0.617240611 -0.060125359 -0.31716015
## BERNARD      0.40062867 -0.40643754  0.703856040  0.170083313 -0.09908142
## YURKOV      -0.10286344 -0.32487448  0.114996135 -0.109524039 -0.11969720
## ZSIVOCZKY   -0.89554111  0.08825624 -0.202341299 -0.523103099 -0.34842265
## McMULLEN    -0.18657283  0.47828432  0.293089967 -0.105623196 -0.39317797
## MARTINEAU    0.45666350 -0.29975522 -0.291628488 -0.223417655 -0.61640509
## HERNU       -0.43623496 -0.56609980 -1.529404317  0.006184409  0.55368016
## BARRAS      -0.09026010  0.21594095  0.682583078 -0.669282042  0.53085420
## NOOL        -0.49716399 -0.53205687 -0.433385655 -0.115777808 -0.09622142
## BOURGUIGNON  0.08200220 -0.59833739  0.563619921  0.525814030  0.05855882
## Sebrle       0.08279306  0.01016177 -0.030585843 -0.847210682  0.21970353
## Clay        -0.77715960  0.25750851 -0.580638301  0.409776590 -0.61601933
## Karpov       0.32958134 -1.36365568  0.345306381  0.193055107  0.21721852
## Macey        0.19728082 -0.26927772 -0.363219506  0.368260269  0.21249474
## Warners      0.55673300 -0.26739400 -0.109470797  0.180283071  0.24208420
## Zsivoczky    0.11383190  0.03991159  0.538039776  0.585966156 -0.14271715
## Hernu        0.33168404  0.44308686  0.247293566  0.066908586 -0.20868256
## Bernard      0.32806564  0.36357920  0.006165316  0.279488675  0.32067773
## Schwarzl     0.68891640  0.56568604 -0.687053339 -0.008358849 -0.30211546
```

```
## Pogorelov     1.59379599  0.78370119 -0.037623661 -0.130531397 -0.03697576
## Schoenbeck  -0.16172381  0.85752368 -0.255850722  0.564222295  0.29680481
## Barras      -1.17415412  0.94512710  0.365550568  0.102255763  0.61186706
```

```r
res.ind$contrib        # Contributions to the PCs
```

```
##                   Dim.1      Dim.2        Dim.3       Dim.4        Dim.5
## SEBRLE       0.03854254  5.7118249 1.385418e+00  0.03572215 8.091161e+00
## CLAY         0.65814114 13.8541889 6.460097e+00  8.55568792 4.034555e+00
## BERNARD      1.86273218  6.1441319 1.349983e-01 19.57827284 4.202070e-02
## YURKOV       0.79686310  0.4431309 2.147558e+01  2.57939100 9.859373e-01
## ZSIVOCZKY    0.01178829  9.6816398 5.974848e+00  0.05231437 2.405750e-01
## McMULLEN     0.01481737  2.3253860 2.496789e+00  9.13531719 1.558646e+01
## MARTINEAU    6.03367104  3.9044125 2.830527e+00  0.73858431 3.118936e+01
## HERNU        3.76996156  3.2842176 8.583863e-02  4.21505626 9.955149e-02
## BARRAS       3.31942012  0.4024544 1.519980e+00  0.27758505 3.388731e-01
## NOOL         8.12988880  5.8489726 1.293851e+00 12.83761115 1.257025e+01
## BOURGUIGNON 20.53729577  3.7757623 3.534995e-04  2.35877858 1.088816e+00
## Sebrle      12.55838616  3.5020697 9.881482e+00  3.47006223 3.889859e-03
## Clay        11.59361384  6.2315181 1.341828e+00  0.53250375 2.007648e+00
## Karpov      17.46609555  1.5072627 3.614914e+00  0.52101693 3.940874e+00
## Macey        3.60207087 10.0732890 3.364879e+00  1.81387486 2.885677e-07
## Warners      2.02910262  0.1879390 1.403071e+01  5.51585696 1.018550e-02
## Zsivoczky    0.23441891  2.0310492 1.047894e+01  0.18126182 1.028128e+00
## Hernu        0.08048777  3.2136178 1.020764e-01  3.82170515 2.946148e+00
## Bernard      1.97075488  5.1661961 2.425213e+00  2.94737426 4.581507e+00
## Schwarzl     0.53184785  1.0252129 3.800546e+00  4.56119277 5.167449e-01
## Pogorelov    0.04843819  1.7533304 3.107757e-01  9.75034337 1.404313e-03
## Schoenbeck   0.25864068  3.9969003 3.722687e-02  0.32970059 2.580092e+00
## Barras       0.10519467  1.5876667 2.605305e+00  1.84296038 3.767994e+00
##                   Dim.6       Dim.7        Dim.8        Dim.9      Dim.10
## SEBRLE       2.21256620 0.621426384 2.992045e-02 12.177477305  0.03819185
## CLAY        17.48801877 0.651413899 6.035125e+00  0.101262442  3.58568943
## BERNARD      1.65019840 2.373652810 7.847747e+00  0.810319793  0.34994507
## YURKOV       0.10878629 1.516564073 2.094806e-01  0.336009790  0.51072064
## ZSIVOCZKY    8.24561722 0.111923276 6.485544e-01  7.664919832  4.32741147
## McMULLEN     0.35788945 3.287016354 1.360753e+00  0.312501167  5.51053518
## MARTINEAU    2.14409841 1.291109482 1.347216e+00  1.398195851 13.54402896
## HERNU        1.95655942 4.604850849 3.705288e+01  0.001071345 10.92781554
## BARRAS       0.08376135 0.670038259 7.380544e+00 12.547331617 10.04537028
## NOOL         2.54127369 4.067669683 2.975270e+00  0.375477289  0.33003418
## BOURGUIGNON  0.06913582 5.144247534 5.032108e+00  7.744571086  0.12223626
## Sebrle       0.07047579 0.001483775 1.481898e-02 20.105546253  1.72063803
## Clay         6.20972751 0.952824148 5.340583e+00  4.703566841 13.52708188
```

```
## Karpov       1.11680500 26.720158115 1.888802e+00  1.043988269  1.68193477
## Macey        0.40014909  1.041910483 2.089853e+00  3.798767930  1.60957713
## Warners      3.18673563  1.027384225 1.898339e-01  0.910422384  2.08904756
## Zsivoczky    0.13322327  0.022889042 4.585705e+00  9.617852173  0.72605208
## Hernu        1.13110069  2.821027418 9.687304e-01  0.125399768  1.55234328
## Bernard      1.10655655  1.899449022 6.021268e-04  2.188071254  3.66566729
## Schwarzl     4.87961053  4.598122119 7.477531e+00  0.001957159  3.25357879
## Pogorelov   26.11665608  8.825322559 2.242329e-02  0.477268755  0.04873597
## Schoenbeck   0.26890572 10.566272800 1.036933e+00  8.917302863  3.14020004
## Barras      14.17432302 12.835417603 2.116763e+00  0.292892746 13.34533825
```

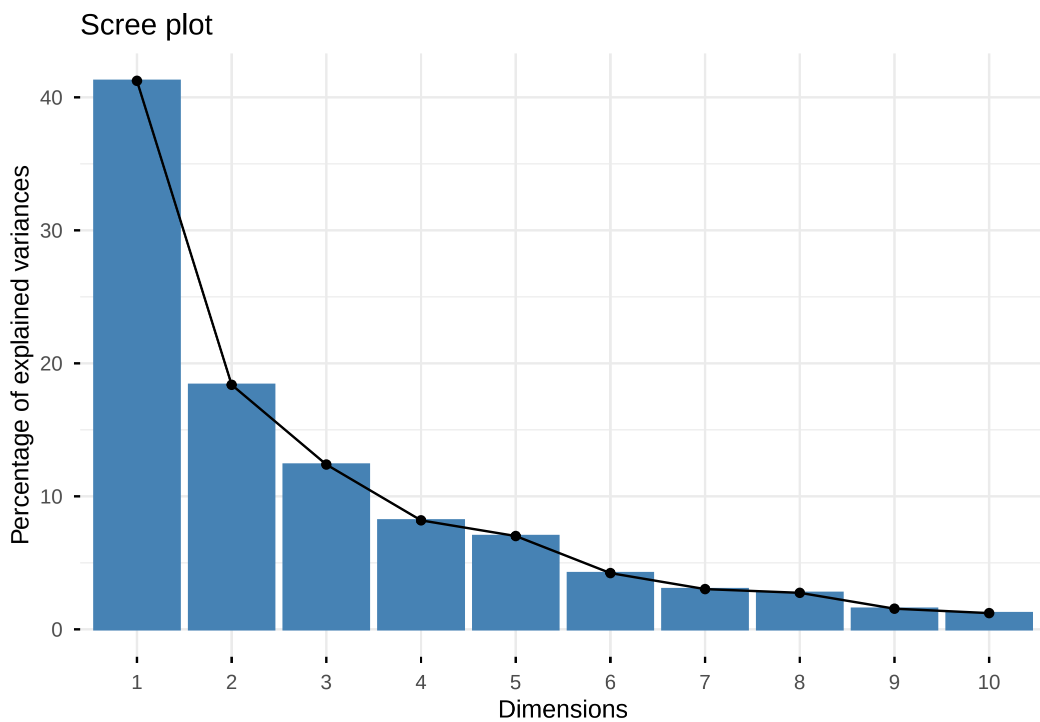
```r
res.ind$cos2            # Quality of representation
```

```
##                   Dim.1      Dim.2        Dim.3       Dim.4        Dim.5
## SEBRLE      0.007530179 0.49747323 8.132523e-02 0.001386688 2.689027e-01
## CLAY        0.048701249 0.45701660 1.436281e-01 0.125791741 5.078506e-02
## BERNARD     0.197199804 0.28996555 4.294015e-03 0.411819183 7.567259e-04
## YURKOV      0.096109800 0.02382571 7.782303e-01 0.061812637 2.022798e-02
## ZSIVOCZKY   0.001574385 0.57641944 2.397542e-01 0.001388216 5.465497e-03
## McMULLEN    0.002175437 0.15219499 1.101379e-01 0.266486530 3.892621e-01
## MARTINEAU   0.404013915 0.11654676 5.694575e-02 0.009826320 3.552552e-01
## HERNU       0.399282749 0.15506199 2.731529e-03 0.088699901 1.793538e-03
## BARRAS      0.616241975 0.03330700 8.478249e-02 0.010239088 1.070152e-02
## NOOL        0.489872515 0.15711146 2.342405e-02 0.153694675 1.288433e-01
## BOURGUIGNON 0.859698130 0.07045912 4.446015e-06 0.019618511 7.753120e-03
## Sebrle      0.675380606 0.08395940 1.596674e-01 0.037079012 3.558507e-05
## Clay        0.687592867 0.16475409 2.391051e-02 0.006274965 2.025440e-02
## Karpov      0.783666922 0.03014772 4.873187e-02 0.004644764 3.007790e-02
## Macey       0.363436037 0.45308203 1.020057e-01 0.036362957 4.952707e-09
## Warners     0.255651956 0.01055582 5.311341e-01 0.138081100 2.182965e-04
## Zsivoczky   0.045053176 0.17401353 6.051030e-01 0.006921739 3.361236e-02
## Hernu       0.024824321 0.44184663 9.459148e-03 0.234196727 1.545686e-01
## Bernard     0.289347476 0.33813318 1.069834e-01 0.085980212 1.144234e-01
## Schwarzl    0.116721435 0.10030142 2.506043e-01 0.198892209 1.929118e-02
## Pogorelov   0.007803472 0.12591966 1.504272e-02 0.312101619 3.848427e-05
## Schoenbeck  0.067070098 0.46204603 2.900467e-03 0.016987442 1.138116e-01
## Barras      0.018972684 0.12765099 1.411800e-01 0.066043061 1.156018e-01
##                    Dim.6        Dim.7        Dim.8        Dim.9       Dim.10
## SEBRLE      0.0443241299 8.907507e-03 3.890334e-04 8.954067e-02 0.0002206741
## CLAY        0.1326907339 3.536548e-03 2.972084e-02 2.820119e-04 0.0078471026
## BERNARD     0.0179131165 1.843634e-02 5.529104e-02 3.228572e-03 0.0010956493
## YURKOV      0.0013453555 1.341980e-02 1.681440e-03 1.525225e-03 0.0018217256
## ZSIVOCZKY   0.1129176906 1.096685e-03 5.764478e-03 3.852703e-02 0.0170924251
## McMULLEN    0.0053876990 3.540616e-02 1.329562e-02 1.726733e-03 0.0239268142
```

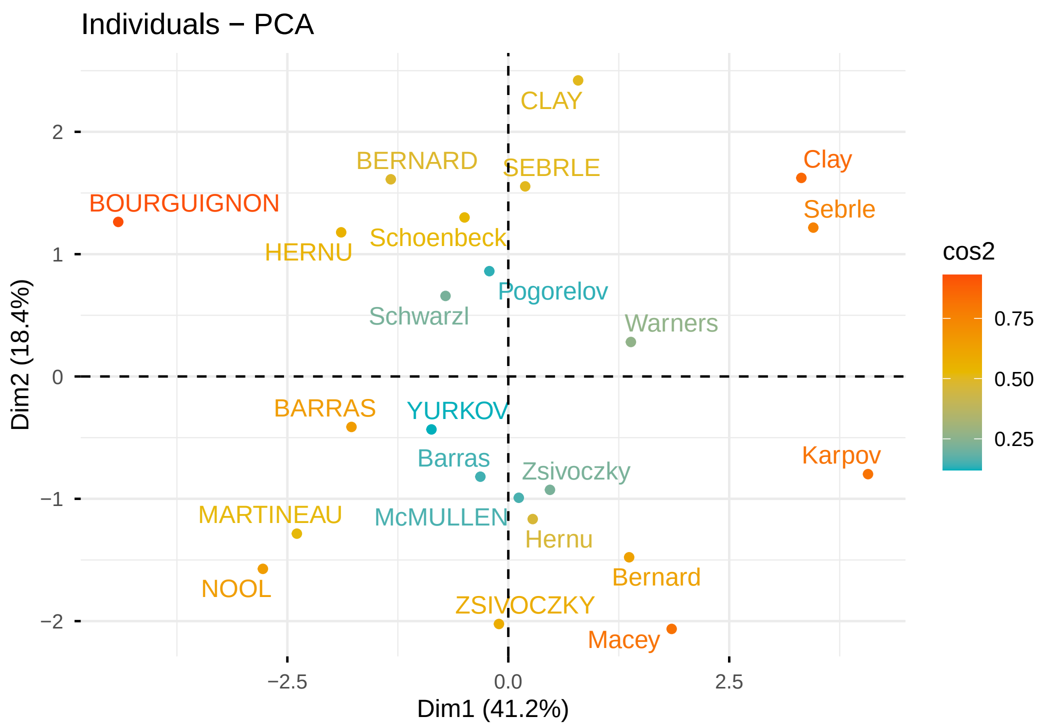
```
## MARTINEAU   0.0147210347 6.342774e-03 6.003515e-03 3.523552e-03 0.0268211980
## HERNU       0.0212478795 3.578167e-02 2.611676e-01 4.270425e-06 0.0342288717
## BARRAS      0.0015944528 9.126203e-03 9.118662e-02 8.766746e-02 0.0551531863
## NOOL        0.0157010551 1.798232e-02 1.193105e-02 8.514912e-04 0.0005881295
## BOURGUIGNON 0.0002967459 1.579887e-02 1.401866e-02 1.220108e-02 0.0001513277
## Sebrle      0.0003886276 5.854423e-06 5.303795e-05 4.069384e-02 0.0027366539
## Clay        0.0377627839 4.145976e-03 2.107924e-02 1.049876e-02 0.0237264222
## Karpov      0.0051379747 8.795817e-02 5.639959e-03 1.762907e-03 0.0022318265
## Macey       0.0041397727 7.712721e-03 1.403282e-02 1.442502e-02 0.0048028954
## Warners     0.0411689767 9.496848e-03 1.591742e-03 4.317040e-03 0.0077841113
## Zsivoczky   0.0026253777 3.227467e-04 5.865332e-02 6.956790e-02 0.0041268259
## Hernu       0.0357707217 6.383462e-02 1.988402e-02 1.455601e-03 0.0141595965
## Bernard     0.0166586433 2.046050e-02 5.883405e-06 1.209056e-02 0.0159167991
## Schwarzl    0.1098063093 7.403638e-02 1.092132e-01 1.616543e-05 0.0211173850
## Pogorelov   0.4314162233 1.043115e-01 2.404103e-04 2.893750e-03 0.0002322016
## Schoenbeck  0.0071500829 2.010275e-01 1.789520e-02 8.702893e-02 0.0240826922
## Barras      0.2621297474 1.698426e-01 2.540745e-02 1.988116e-03 0.0711836486
```

Visualizations of PCA results

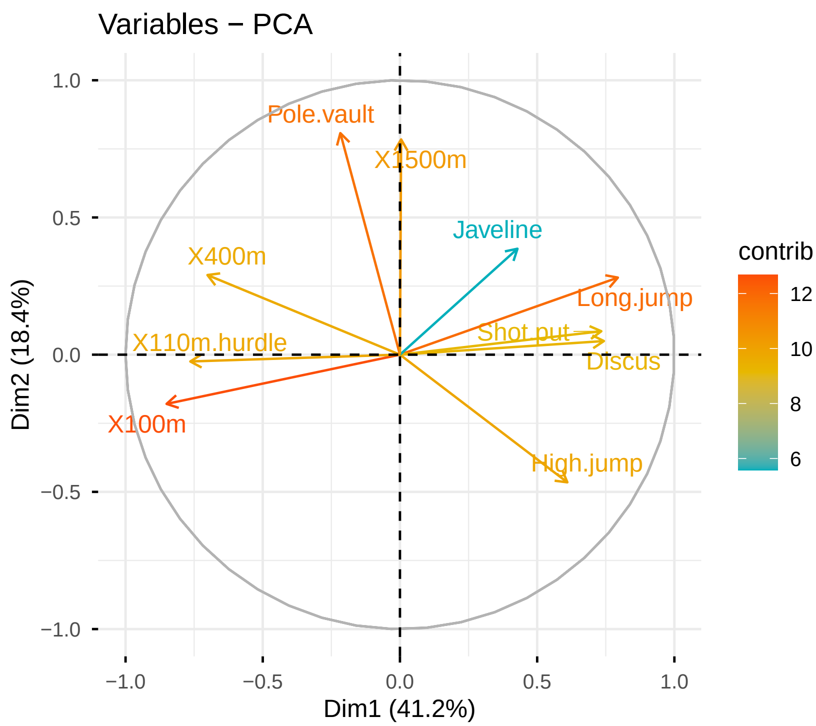
```
#Visualize eigenvalues (scree plot). Show the percentage of variances
#explained by each principal component.
fviz_eig(res.pca)
```




```
#Graph of individuals. Individuals with a similar profile are grouped together.
fviz_pca_ind(res.pca,
             col.ind = "cos2", # Color by the quality of representation
             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
             repel = TRUE     # Avoid text overlapping
             )
```




```
#Graph of variables. Positive correlated variables point to the same side of
#the plot. Negative correlated variables point to opposite sides of the graph.
fviz_pca_var(res.pca,
             col.var = "contrib", # Color by contributions to the PC
             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
             repel = TRUE     # Avoid text overlapping
             )
```




```
#Biplot of individuals and variables
fviz_pca_biplot(res.pca, repel = TRUE,
                col.var = "#2E9FDF", # Variables color
                col.ind = "#696969"  # Individuals color
                )
```